StaxED

The Syntax of Physical Reality

Units, Dimensions, and the Integrity of Measurement

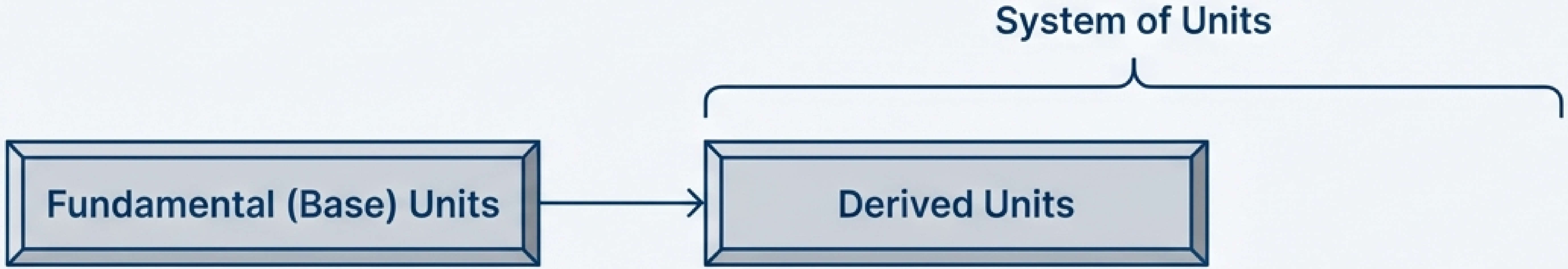
To describe the universe, we must first agree on the language. Measurement is not just data collection; it is the syntax we use to compare physical reality against universal standards.

Measurement is Comparison

Measurement is the process of comparing a physical quantity against an internationally accepted reference standard known as a Unit.

How many times? *The Standard*

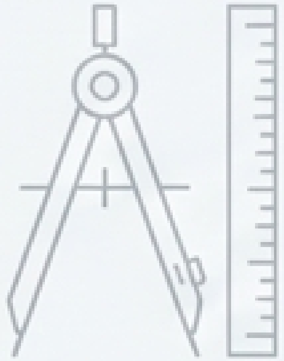
$$\text{Physical Quantity} = \text{Numerical Value} \times \text{Unit}$$



Independent building blocks (Length, Mass, Time)

Combinations of base units (Speed, Density)

The Evolution of Standards



Historic Chaos

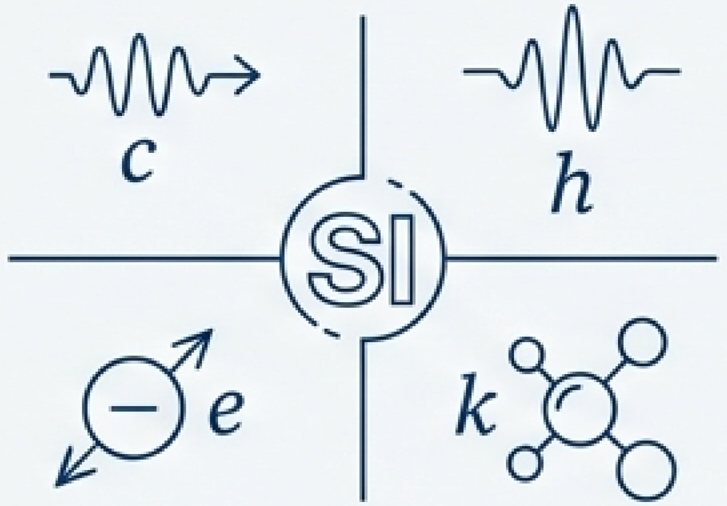
CGS, FPS, MKS systems.
Regionally dependent.

1971

SI Adopted

The International Syster
of Units establishes a
global decimal standard

Why SI? It is a Base-10 decimal system, simplifying global scientific communication.

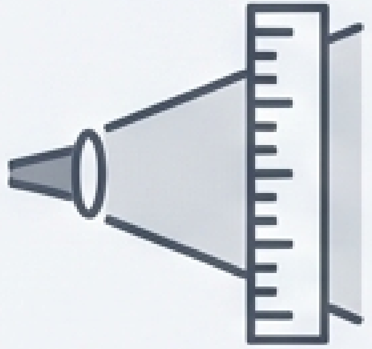


2018 Revision

The Quantum Shift

Definitions decoupled from artifacts and anchored to fixed universal constants (c, h, e, k).

The Seven Pillars of SI

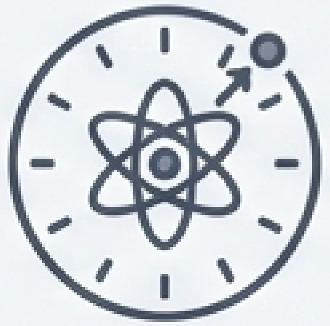


Metre (m)

Defined by speed of light c.

Kilogram (kg)

Defined by Planck constant h.



Second (s)

Defined by Caesium frequency.

Ampere (A)

Defined by elementary charge e.

Kelvin (K)

Defined by Boltzmann constant k.

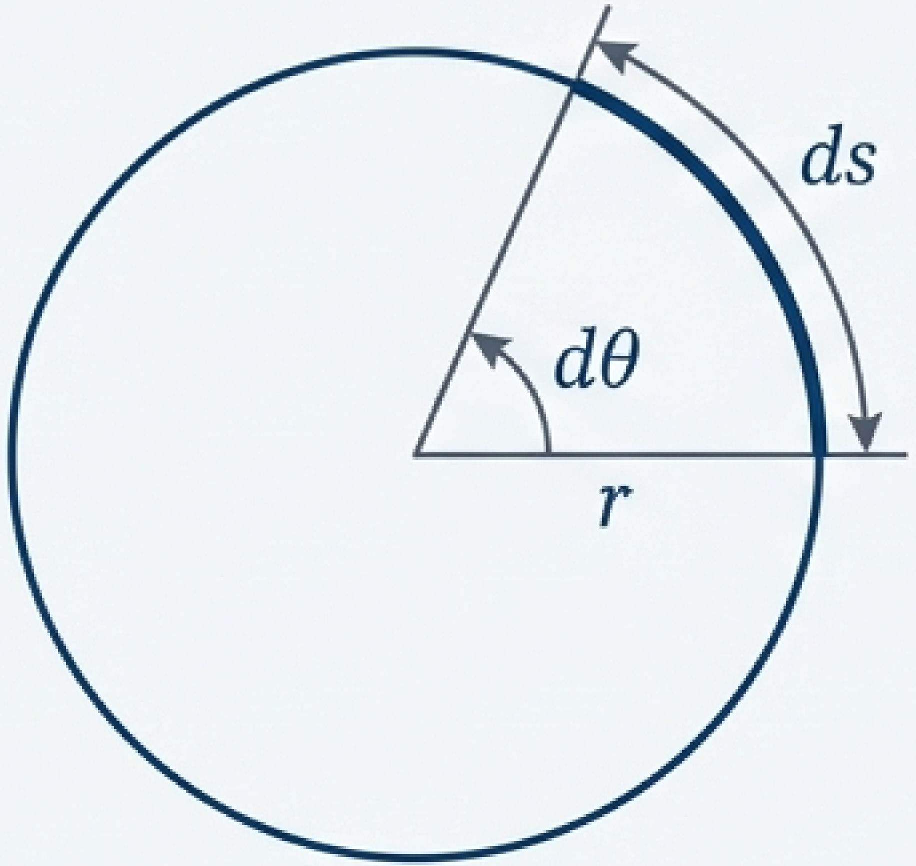
Mole (mol)

Defined by Avogadro constant.

Candela (cd)

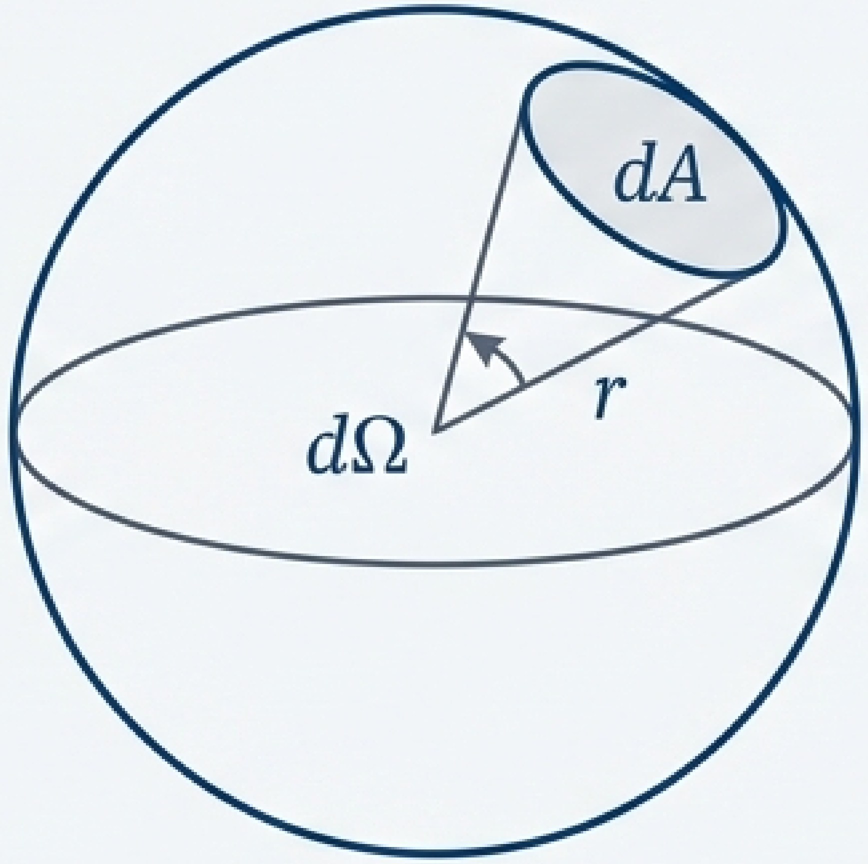
Defined by luminous efficacy.

Supplementary and General Use Units



Radian (rad): Plane angle.

$$\mathrm{d}\theta = \frac{ds}{r}$$



Steradian (sr): Solid angle.

$$\mathrm{d}\Omega = \frac{dA}{r^2}$$

Crucial: These are Dimensionless quantities.

General Use (Non-SI)

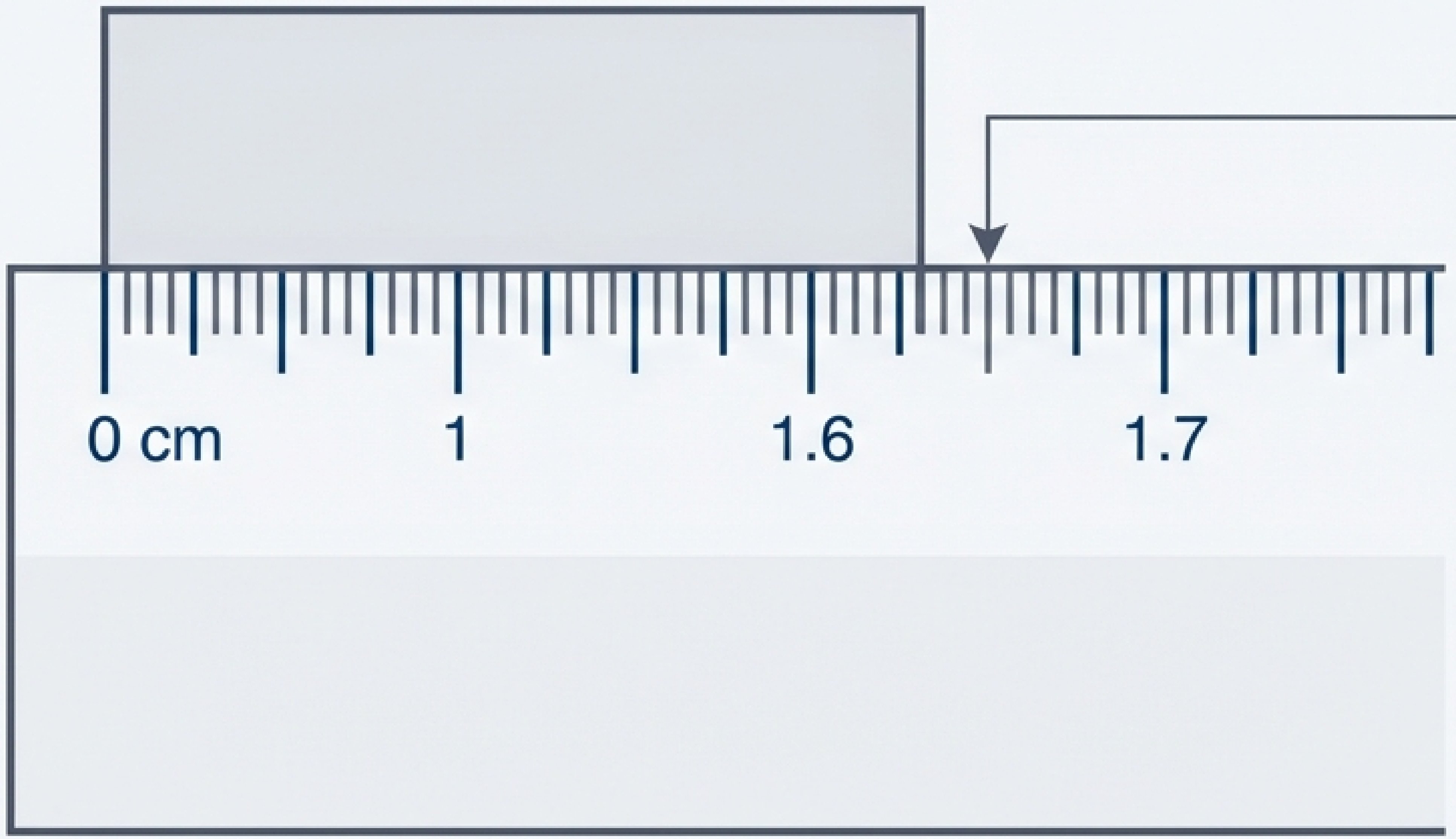
Time: minute, hour, day

Volume: litre (1 dm^3)

Mass: tonne (1000 kg)

Pressure: bar (10^5 Pa)

Significant Figures: The Art of Precision



Significant figures consist of all reliably known digits plus the first uncertain digit. They indicate the precision of the measurement.

Changing units (e.g., 1.62 s to 1620 ms) does **NOT** change the number of significant figures.

Rules of Significance

Trapped Zeros in Technical Slate		Leading Zeros in IBM Plex Serif	
2.05	Significant	0.0023	Not Significant in Technical Slate
Trailing Zeros (No Decimal) in Technical Slate	Not Significant in Technical Slate	12300	Not Significant in Technical Slate
Trailing Zeros (With Decimal) in Technical Slate	Significant in Technical Slate	3.500	Significant
Scientific Notation in Technical Slate	4.70×10^3 Only digits in the base number count. in IBM Plex Serif		

Calculation Integrity

Multiplication / Division

Result is limited by the **LEAST** significant figures.

$$\text{Density} = 4.237\ \text{g} \div 2.51\ \text{cm}^3$$
$$= 1.68804\ldots$$
$$= 1.69\ \text{g cm}^{-3}$$

Rounded to 3 sig figs (matching 2.51)

Addition / Subtraction

Result is limited by the **LEAST** decimal places.

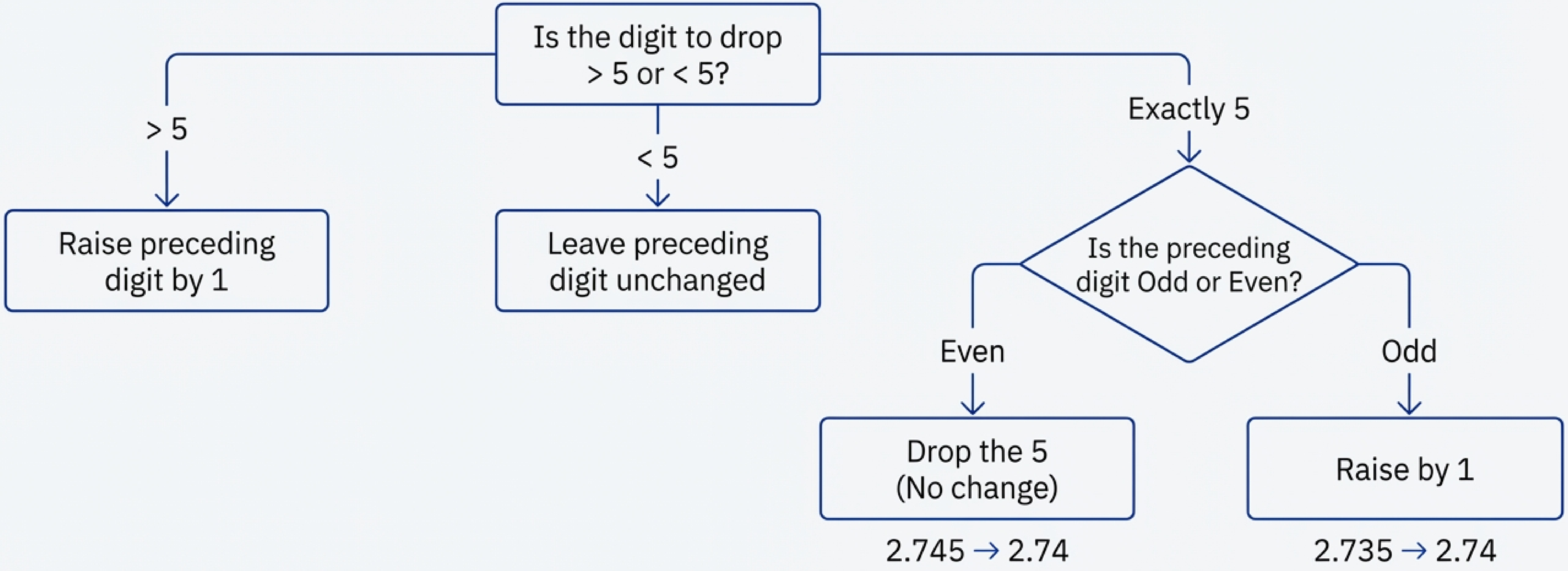
$$\begin{array}{lr} & 436.32 \\ + & 227.2 \\ + & 0.301 \\ \hline = & 663.821 \end{array}$$

$$663.8$$

Rounded to 1 decimal place

NotebookLM

Rules for Rounding Uncertain Digits



Avoid cumulative error: Retain one extra digit during intermediate steps and round only at the end.

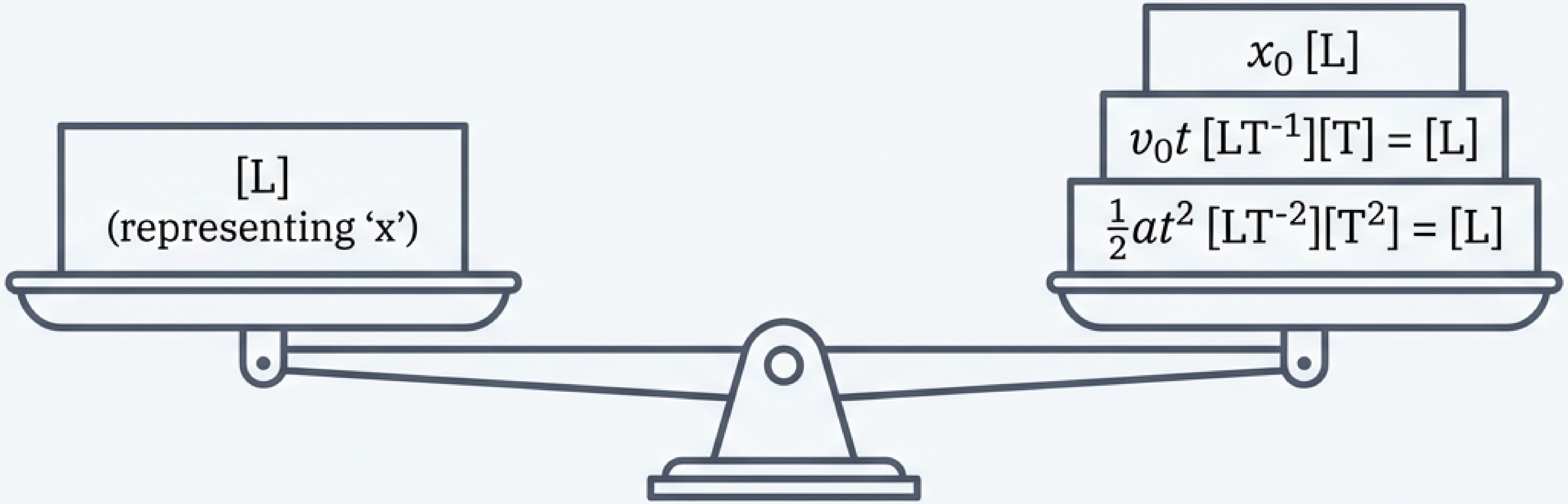
Dimensions: The DNA of Physics

Dimensions describe the nature of a quantity, ignoring magnitude.
Notation: [L] Length, [M] Mass, [T] Time.

Standard Formulas	Dimensional Translation
Volume: Length × Width × Height	$[\mathbf{L}^3]$
Speed: Distance / Time	$[\mathbf{LT}^{-1}]$
Force: Mass × Acceleration	$[\mathbf{MLT}^{-2}]$
Density: Mass / Volume	$[\mathbf{ML}^{-3}]$

The Principle of Homogeneity

You cannot add Apples to Oranges.



$$x = x_0 + v_0 t + \frac{1}{2}at^2$$

Dimensionally Correct.

Side note: Special functions (sin, log, e^x) must be dimensionless.

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Application: Deducing Relationships

Derive the Time Period (T) of a pendulum. We know it depends on Length (l), Mass (m), and Gravity (g).

$$T = k \cdot l^x \cdot g^y \cdot m^z$$

$$[T^1] = [L]^x \cdot [LT^{-2}]^y \cdot [M]^z$$

Group terms:

$$[L^0 M^0 T^1] = L^{x+y} \cdot T^{-2y} \cdot M^z$$

1. $z = 0$

   (Mass is irrelevant)

2. $-2y = 1 \rightarrow y = -\frac{1}{2}$

3. $x + y = 0 \rightarrow x = \frac{1}{2}$

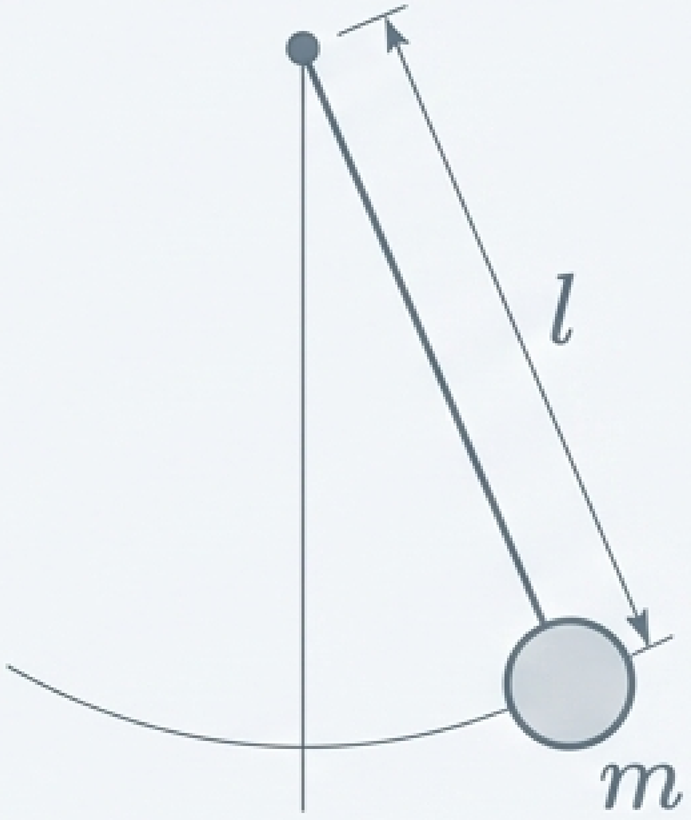
$$\boxed{T = k \cdot l^{\frac{1}{2}} \cdot g^{-\frac{1}{2}} \cdot m^0}$$

✅ Dimensionally Correct. Final form: $T = k\sqrt{(l/g)}$

The Result & Limitations

$$T = k \cdot l^{\frac{1}{2}} \cdot g^{-\frac{1}{2}}$$

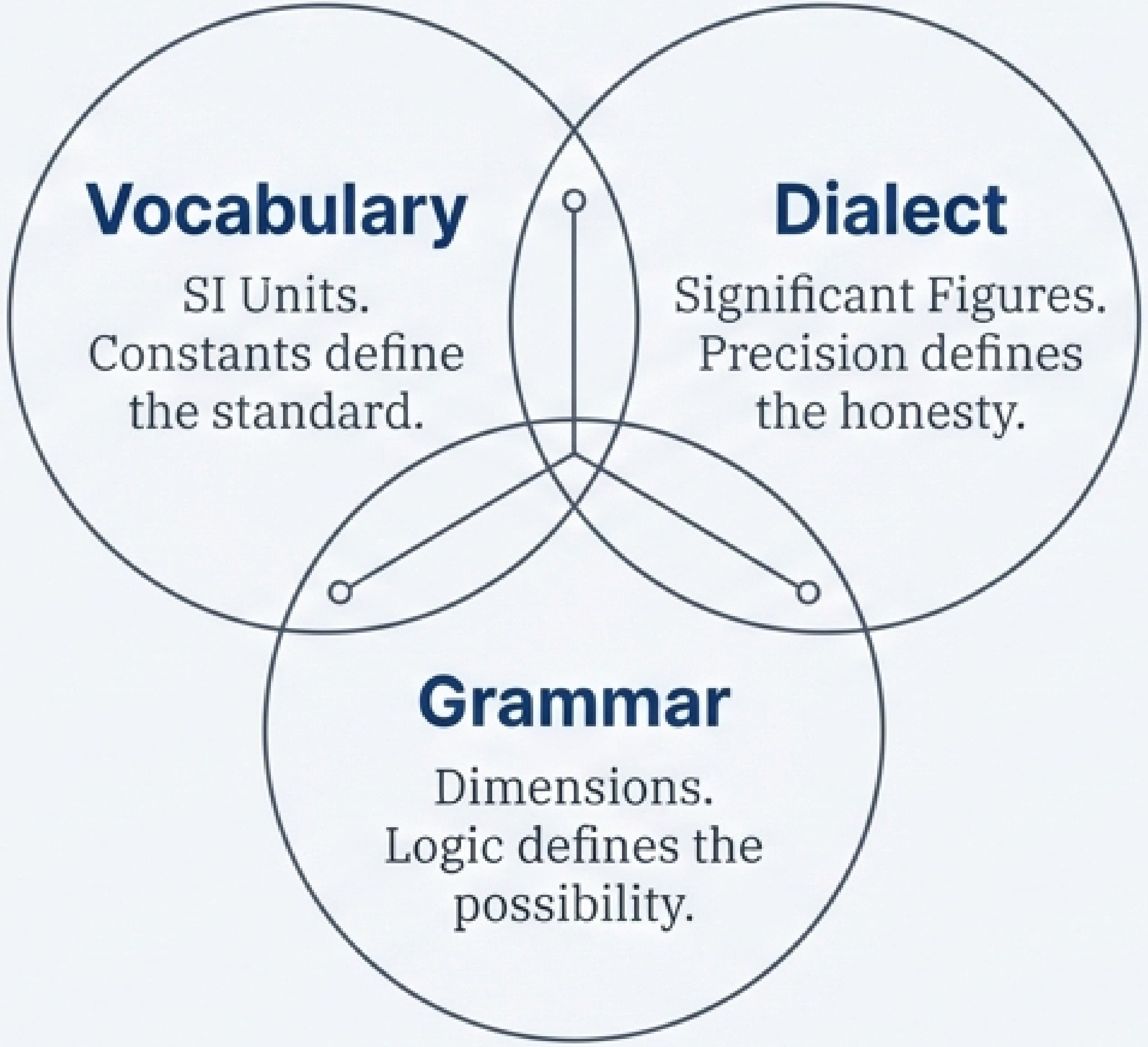
$$T = k\sqrt{\frac{l}{g}}$$



Caution

Dimensional analysis are limited to:
1. Cannot find value of dimensionless constant k (here k=2π).
2. Cannot distinguish between quantities with same dimensions (e.g., Energy vs Torque).

Mastering the Language



"In physics, a number without a unit is meaningless; a unit without precision is misleading; and an equation without dimensional balance is impossible."